

11 (a) minimum frequency for electron(s) to be emitted (from surface)
reference to frequency of electromagnetic radiation / photon

M1
A1

or

frequency causing emission of electron(s)
from surface with zero kinetic energy
reference to frequency of electromagnetic radiation / photon

(M1)
(A1) [2]

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- (b) (i) positive intercept on $(1/\lambda)$ -axis (when extrapolated)
straight line with positive gradient
B1
B1 [2]
- (ii) gradient = hc where c is the speed of light
B1 [1]
- (iii) maximum kinetic energy when electron emitted from surface
energy is required to bring an electron to the surface
B1
B1 [2]
- (iv) each photon has more energy
fewer photons per unit time
fewer electrons per unit time/less current
M1
M1
A1 [3]